Predicting Vortex Shedding using MeshGraphNet Architecture

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Mechanical Science & Engineering

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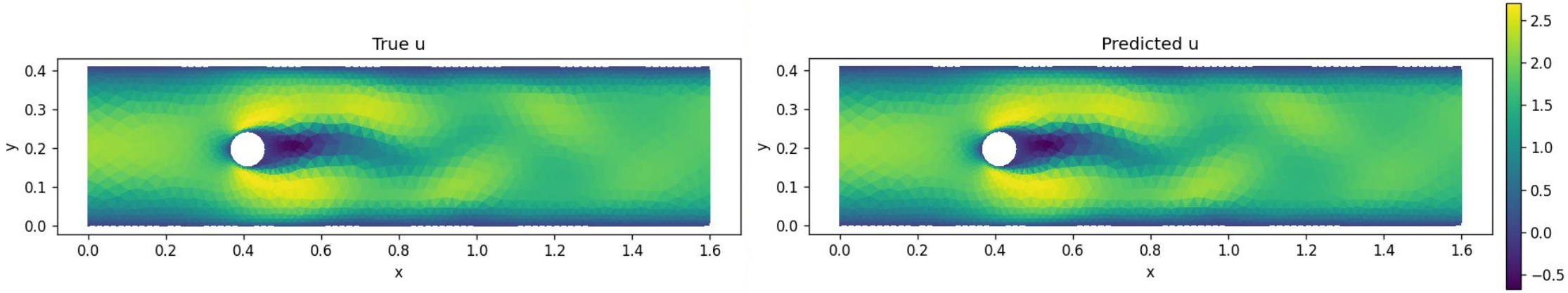
Predict a flow past a cylinder



- **Problem:** predict flow evolution on an unstructured mesh
- **Challenge:** rollout errors can accumulate over time
- **Goal:** implement MeshGraphNet and compare one-step vs rollout behavior
- **Implementation:** full ML-pipeline with logging (wandb) and cluster capabilities

Data + Model

Pfaff *et. al.* 2020

Original Dataset (16Gb):

1200 trajectories w/ 600 time-steps each

1000 train

100 validation

100 test

Nvidia 4060Ti

5 trajectories w/ 600 time-steps each

3 train

1 validation

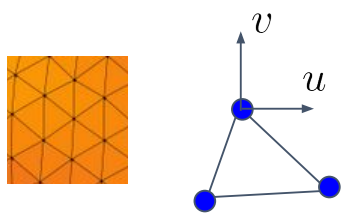
1 test

Residual Connection

Encoder

Processor

Decoder



node and edge embeddings

2 MLPs

$n, e \rightarrow n', e'$

(latent space)

2 MLPs (MP and U)

$$MP = n_1' - e' - n_2' \rightarrow e'^{+1}$$

$$A = \begin{cases} \rightarrow \sum_i e_i'^{+1} \\ \text{stack}[n', \sum_i e_i'^{+1}] \end{cases}$$

$$U = \rightarrow \text{stack}[n', \sum_i e'^{+1}] \rightarrow n'^{+1}$$

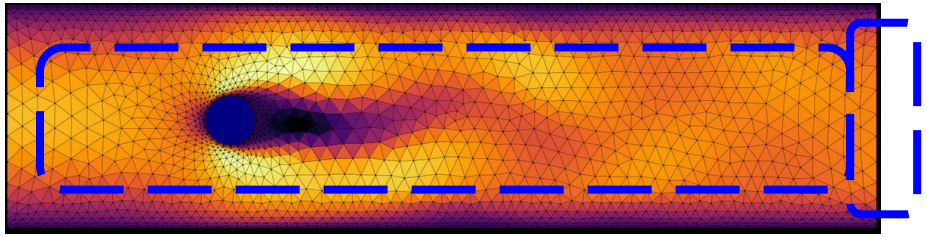
Loss

$$\mathcal{L} = \sqrt{\frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|\tilde{y}_i - \hat{y}_i\|_2^2}$$

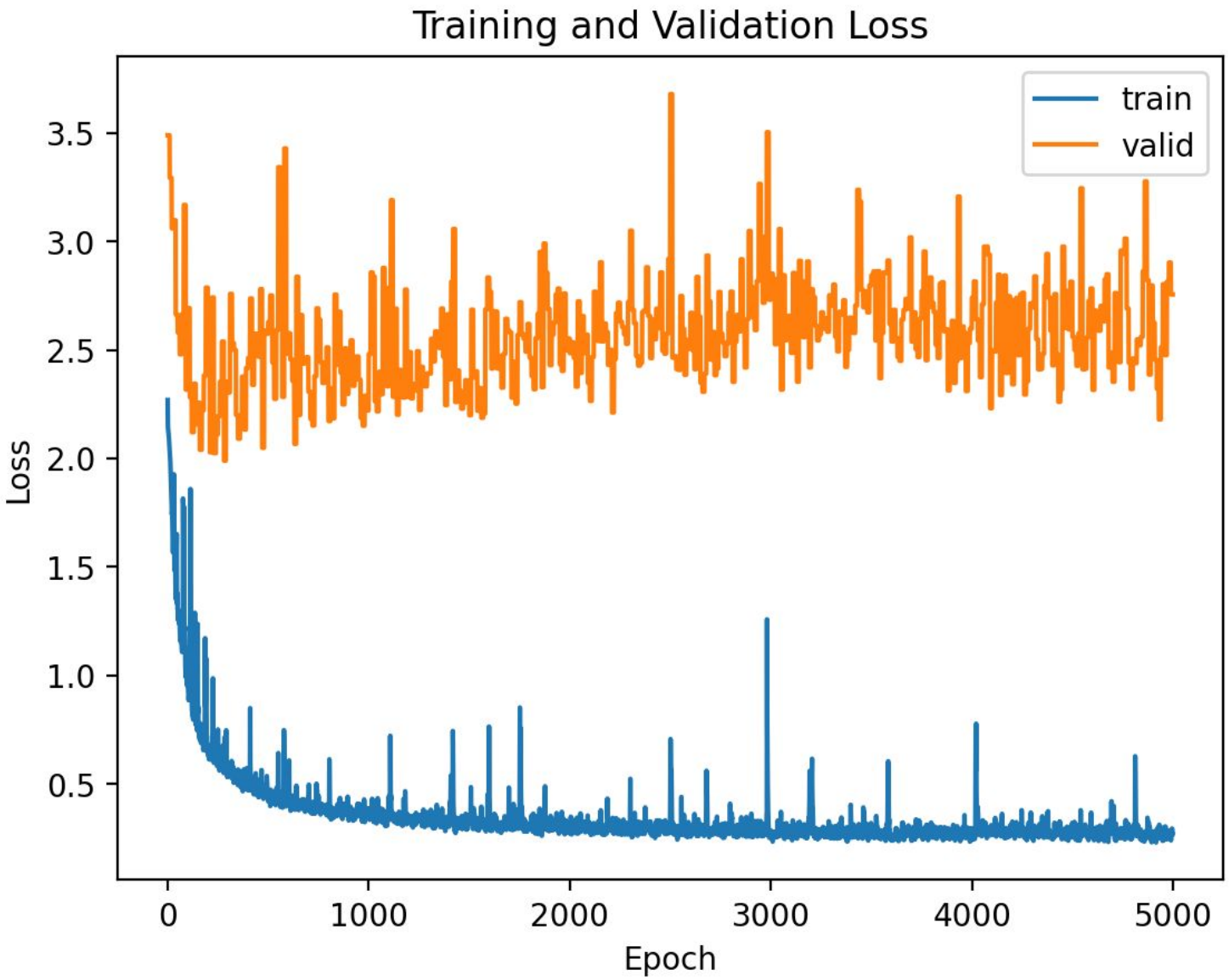
$\mathcal{M}$ is the set of nodes used in the Loss

$\tilde{y}_i$ is the normalized true target at node i

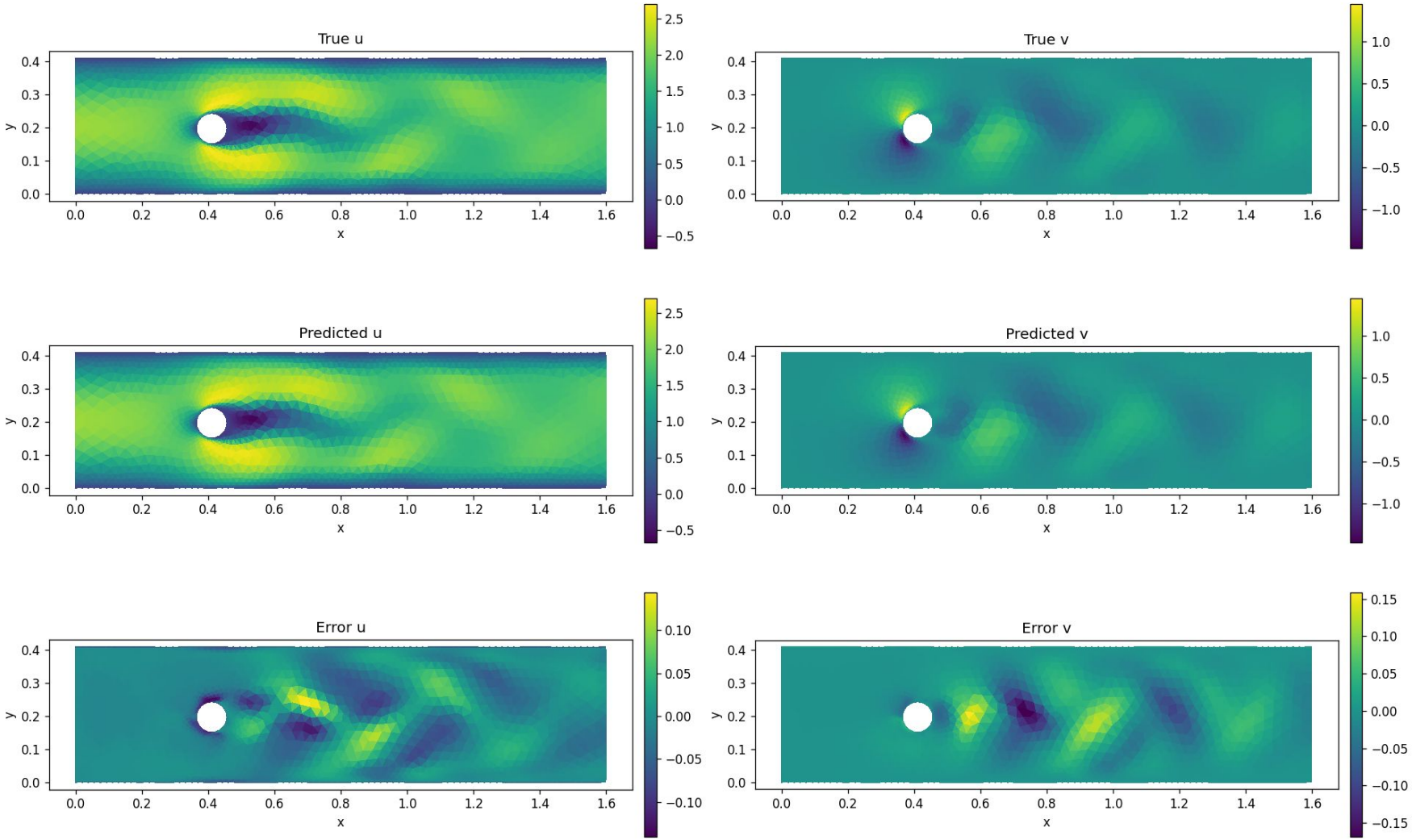
$\hat{y}_i$ is the predicted target at node i

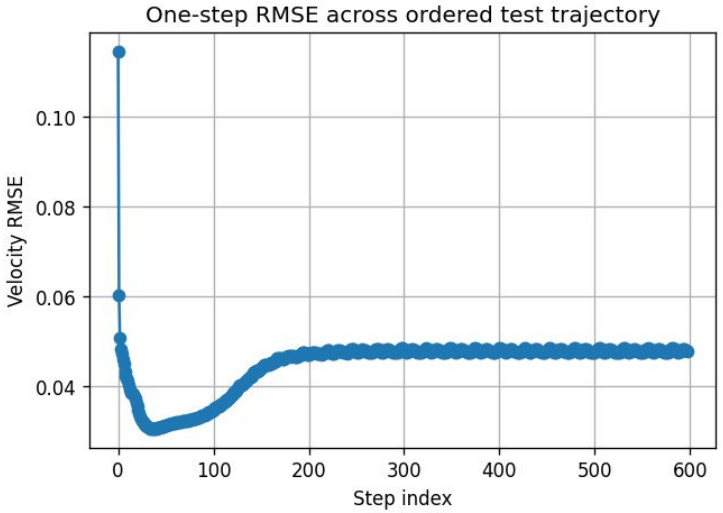


$\mathcal{M} = \text{normal+outflow}$



One-step results

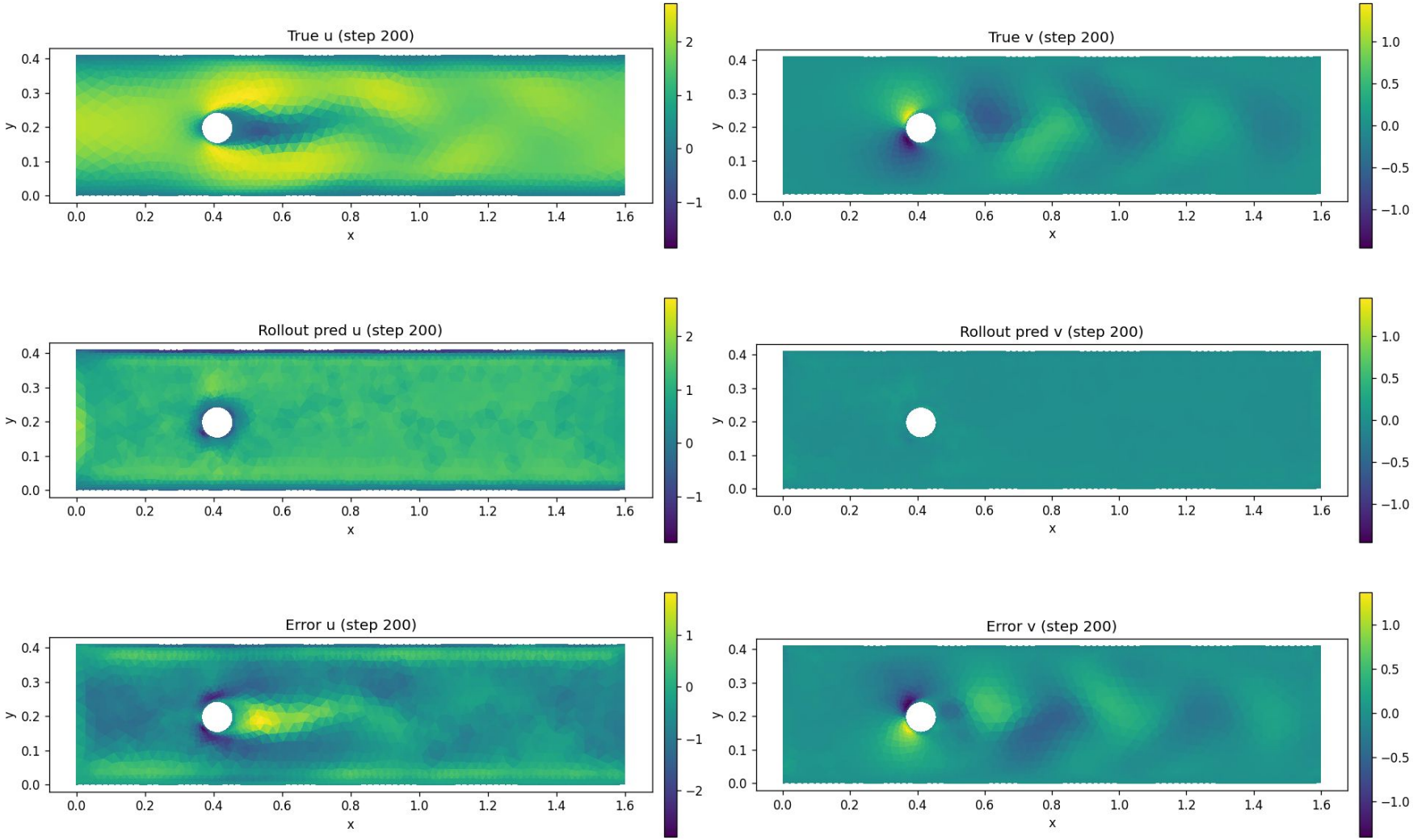




Using test data (1 trajectory)

- The model captures short-term flow evolution on the mesh with relatively low one-step error

Rollout results

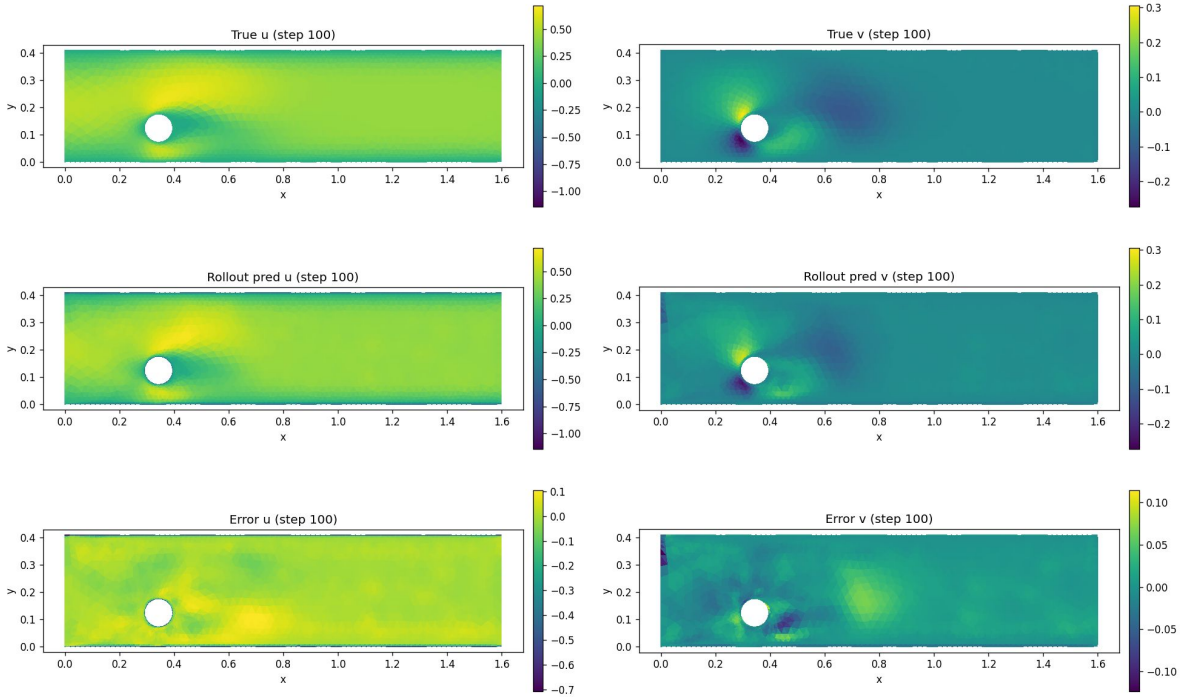


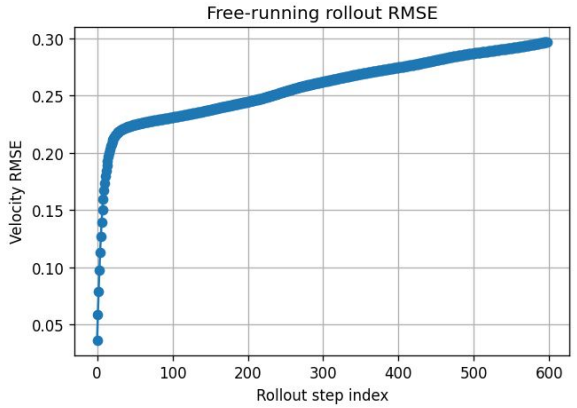
Using test data (1 trajectory)

- one-step success does not guarantee stable rollout
- the main challenge is generalization and compounding error

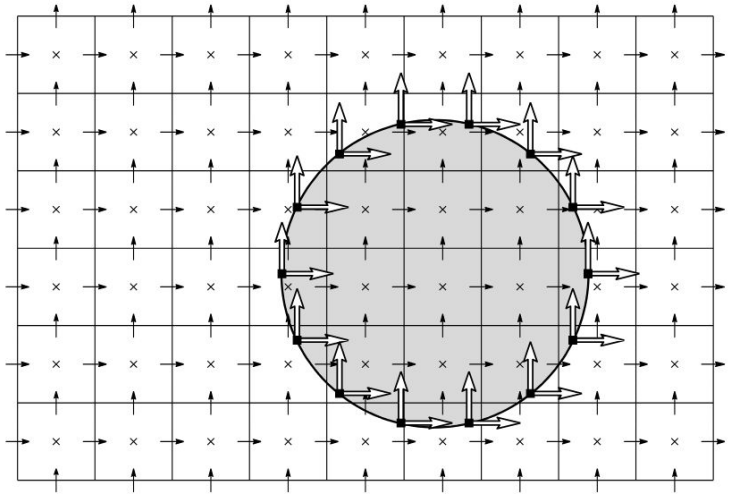
Reflections and next steps

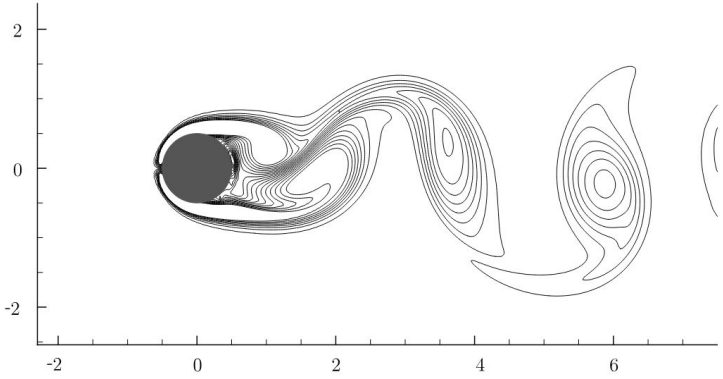
Using train data





Immersed Boundary Method
2 meshes (Eulerian + Lagrangian)





Taira and Colonius (2007)

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References

1. Pfaff, Tobias, et al. "Learning mesh-based simulation with graph networks." arXiv preprint arXiv:2010.03409 (2020).

2. Taira, Kunihiko, and Tim Colonius. "The immersed boundary method: a projection approach." Journal of Computational Physics 225.2 (2007): 2118-2137.

Learning Mesh-Based Simulation with Graph Networks

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Abstract

Mesh-based simulations are central to modeling complex physical systems in many disciplines across science and engineering, as they support powerful numerical integration methods and their resolution can be adapted to strike favorable trade-offs between accuracy and efficiency. Here we introduce MESHGRAPHNETS, a graph neural network-based method for learning simulations, which leverages mesh representations. Our model can be trained to pass messages on a mesh graph and to adapt the mesh discretization during forward simulation. We show that our method can accurately predict the dynamics of a wide range of physical systems, including aerodynamics, structural mechanics, and cloth– and do so efficiently, running 1-2 orders of magnitude faster than the simulation on which it is trained. Our approach broadens the range of problems on which neural network simulators can operate and promises to improve the efficiency of complex, scientific modeling tasks.

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Journal of Computational Physics 225 (2007) 2118–2137

JOURNAL OF COMPUTATIONAL PHYSICS
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The immersed boundary method: A projection approach

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Received 11 April 2006; received in revised form 11 December 2006; accepted 12 March 2007
Available online 19 March 2007

Abstract

A new formulation of the immersed boundary method with a structure algebraically identical to the traditional fractional step method is presented for incompressible flow over bodies with prescribed surface motion. Like previous methods, a boundary force is applied at the immersed surface to satisfy the no-slip constraint. This extra constraint can be added to the incompressible Navier–Stokes equations by introducing regularization and interpolation operators. The current method gives prominence to the role of the boundary force acting as a Lagrange multiplier to satisfy the no-slip condition. This role is analogous to the effect of pressure on the momentum equation to satisfy the divergence-free constraint. The current immersed boundary method removes slip and non-divergence-free components of the velocity field through a projection. The boundary force is determined implicitly without any constitutive relations allowing the present formulation to use larger CFL numbers compared to some past methods. Symmetry and positive-definiteness of the system are preserved such that the conjugate gradient method can be used to solve for the flow field. Examples show that the current formulation achieves second-order temporal accuracy and better than first-order spatial accuracy in L_2-norms for one- and two-dimensional test problems. Results from two-dimensional simulations of flows over stationary and moving cylinders are in good agreement with those from previous experimental and numerical studies.